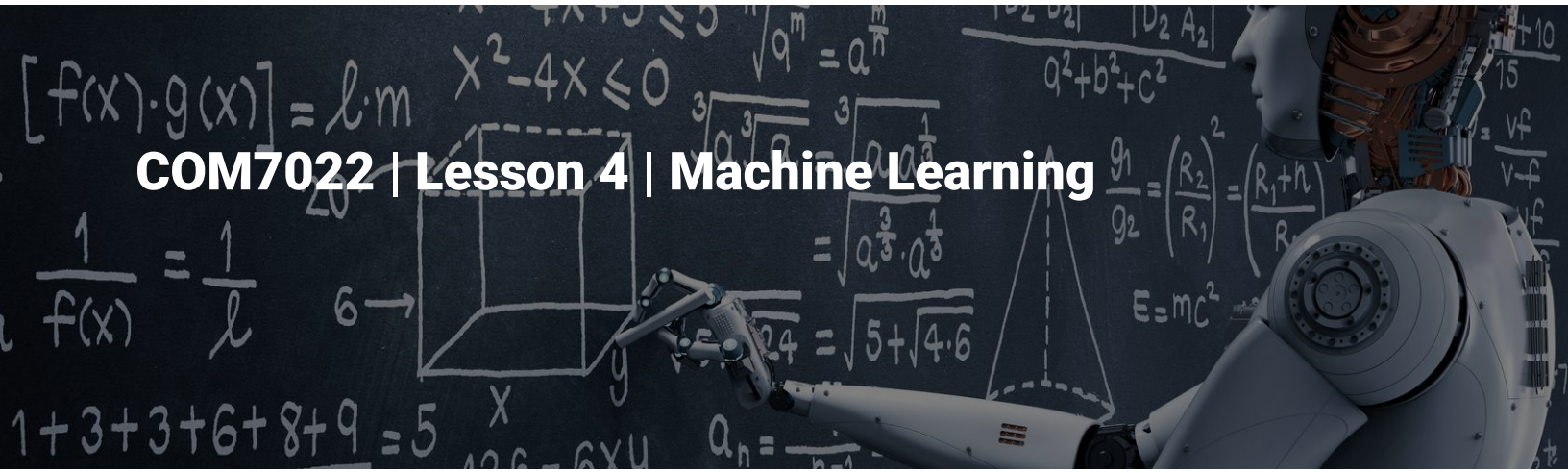


COM7022 | Lesson 4 | Machine Learning

Machine Learning (COM7022)

Lesson 4: Association rules

Cover image by Phonlamai Photo from Shutterstock.com

- **Lesson introduction**
- **Before you get started**
- **Topic 1: Introduction to association rules**
- **Topic 2: The apriori algorithm**
- **Topic 3: Practical implementation of apriori algorithm using Jupyter Notebook**
- **Lesson summary**
- **Knowledge check**

Reading List

Lesson introduction

Welcome to Lesson 4

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

Welcome to Lesson 4!

In this lesson, we will explore the exciting topic of association rules. This is a popular technique for finding the association between items (attributes) within a dataset. In this case, we are not looking at the correlation, but at the frequency of one item A existing within the data, implying that item B also exists in the itemset.

A common use of association rule mining is market basket analysis, where the contents of individual transactions (or shopping baskets) are analysed to discover what products people often buy together. This can then be used to recommend further items. For example, suppose the data mining reveals that there is an association suggesting that people who buy running shoes also buy water bottles. In that case, the retailer may recommend water bottles to future customers who buy running shoes.



Computing

At Arden University

LESSON INTRODUCTION | LEARNING OUTCOMES | ACCESSIBILITY GUIDANCE

By the end of this lesson, you will:

- Understand and be able to find **frequent item sets** from a dataset
- Understand the principle of the **apriori algorithm** and how to implement it onto an itemset
- Know how to apply **rule generation** to new items
- Explore the implementation of the **apriori algorithm** in Jupyter Notebook

Based on the above, Lesson 3 aligns with all three learning objectives:

1. Demonstrate a practical understanding of the characteristics of various machine-learning methods and their applications in data analysis.
2. Select and apply appropriate machine-learning techniques for a given scenario.

3. Critically evaluate the result from the application of machine-learning techniques and propose improvements.

The following are the topics we will cover in this lesson:

- Topic 1: Introduction to association rules
- Topic 2: The apriori algorithm
- Topic 3: Practical implementation of the apriori algorithm using Jupyter Notebook



We have worked across teams at Arden University to try and create an accessible and inclusive experience within the learning content. We are continuously assessing

the accessibility of our lessons to make improvements and enhancements based on testing and feedback.

This lesson may include screenshots of code and software used. Please be aware that these may not always have a text alternative, and that other features such as size and colour may not be accessible.

If you experience any issues with accessing the learning content throughout this module, there are several support services in place to help you.

- Arden University has a dedicated Inclusion Services Team who can help you with getting adjustments and equipment.
- The Library Portal also has information on the accessibility tools that can help you access resources: Accessibility Tools – Library Portal
- If you are struggling with accessing a specific part of the content e.g. a video resource that is missing captions, please speak to your Lecturer who may be able to advise on an alternative resource in the interim to ensure you are able to get the learning you need to progress with the lesson.

Continue to the Before you get started section

Section 2 of 8

Before you get started



Image from Articulate Rise 360 Content Library.

Did you know?

In 2013, what percent of all Amazon sales do you think were generated by recommendations to customers?

- ○ 5%
- ○ 15%
- ○ 25%
- ○ 35%

SUBMIT

Continue

In 2013, it was reported that 35% of all sales on Amazon were estimated to be generated by recommending products to customers (MacKenzie, Meyer and Noble, 2013).

Continue

According to a 2011 study on a Canadian university course, students who frequently participated in discussion forums were less likely to receive low exam scores (Cheng *et al.*, 2011).

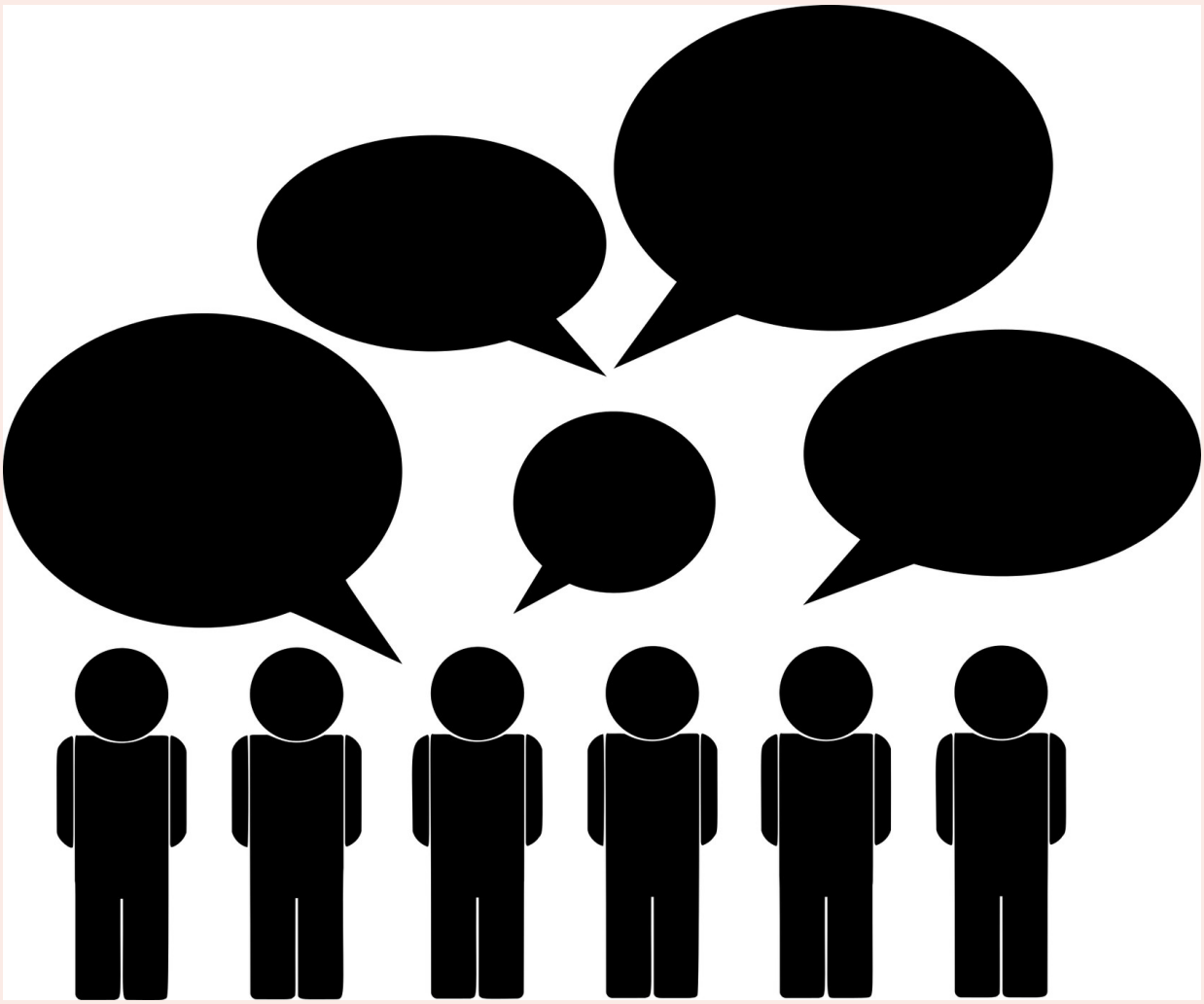


Image from Articulate Rise 360 Content Library.

Continue to Topic 1

Topic 1: Introduction to association rules

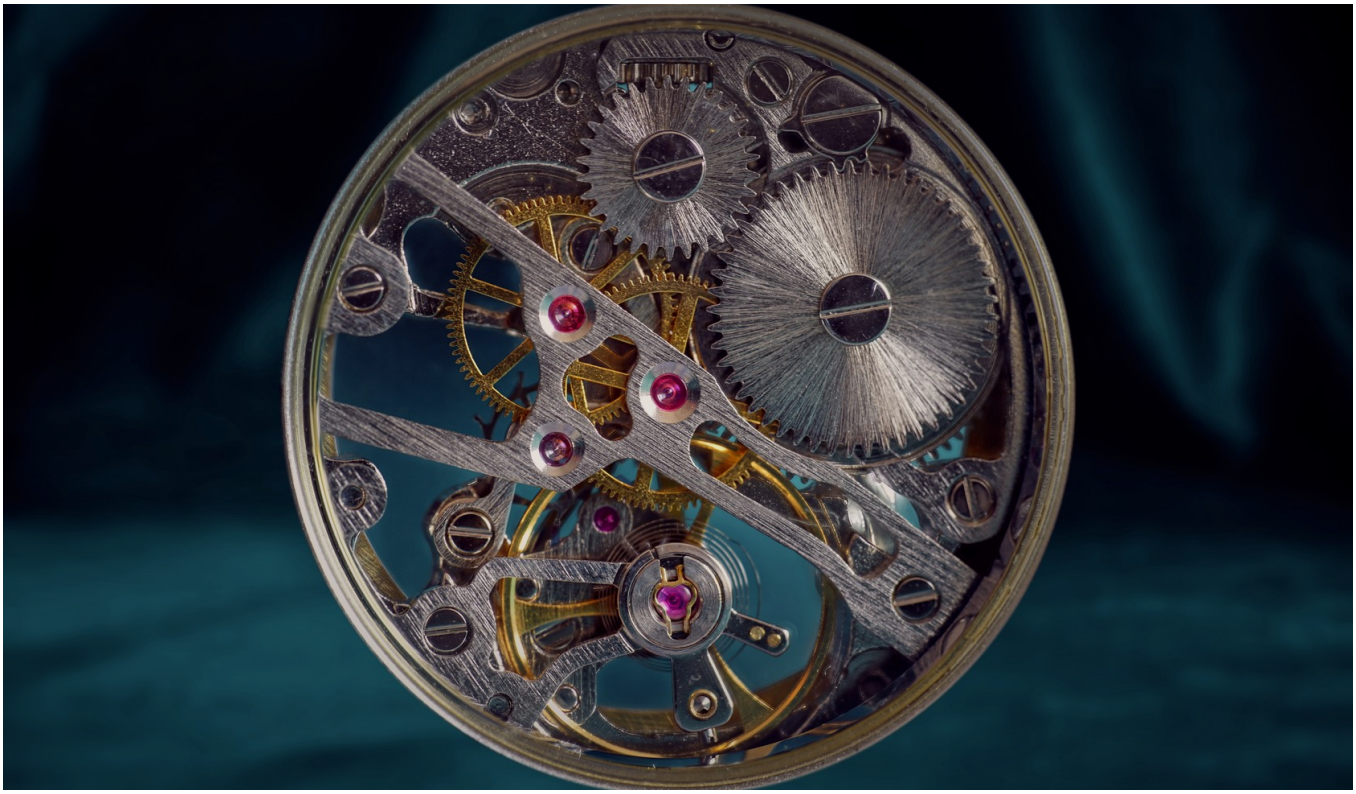


Image from Articulate Rise 360 Content Library.

In this topic, we will formally define an association rule and learn some key terms and metrics used when mining association rules.

To illustrate these terms, we will take the common example of a market basket analysis of transactions at a supermarket. For simplicity, we will assume that the supermarket only sells five

items: bread, milk, flour, eggs and cheese, and our dataset only contains the following four transactions:

Transaction 1: eggs, milk

Transaction 2: eggs, flour, milk

Transaction 3: bread, cheese, eggs, flour

Transaction 4: cheese, eggs, flour, milk

Now let's define some key terms.

An **association rule** between two attributes *A* and *B* in a dataset describes the tendency that if a record has the attribute *A* then it will also have the item *B*. We write the rule as:

$$A \Rightarrow B$$

We read this as the rule "If *A* then *B*". *A* is known as the **antecedent**, and *B* is known as the **consequent**. The attributes in the rule are often described using predicates and variables.

A **predicate** is an operator (often a verb) that is applied to some variables to get a description of an attribute of a record or transaction in our dataset. In the case of a market basket analysis, there is normally only one predicate. In the following example we will use *buys*, so that:

- *buys*(*bread*) - describes the fact that in a transaction the customer has bought bread

- $buys(cheese) \wedge buys(eggs)$ - describes the fact that in a transaction the customer has bought bread

Putting this together, we can now express association rules such as:

$$buys(cheese) \text{ and } buys(eggs) \Rightarrow buys(bread)$$

This rule describes the association that if the customer buys cheese and eggs, they would also buy bread.

When there is only one predicate (such as in a market basket analysis), we can drop the predicate and write the rule in terms using sets of items (or *itemsets*) like this:

$$\{cheese, eggs\} \Rightarrow \{bread\}$$

In this context, we understand what this means without the predicate.

An **itemset** is a group of the items referred to in the dataset. This is normally a list written in braces. In our example, possible item sets are {*bread*}, {*cheese*} or {*bread, eggs*}.

We can classify an itemset by the number of items in it. So {*bread*}, {*cheese*} are know as 1{*bread*, *eggs*} a 2-itemset. In general, we refer to a ***k*-itemset**, where k is a number that tells us how many items are in the itemset.

Now that we know what an association rule is and how it is written, let's look at some key metrics that are used to evaluate association rules and decide which are more interesting to consider and act on. These are support, confidence and lift.

The **support** of an association rule $A \Rightarrow B$ is the proportion of transactions in our dataset that contain both attributes *A* and *B*. This is the probability of *A* and *B* co-occurring in the same transaction:

$$support(A \Rightarrow B) = \frac{Number\ of\ transactions\ containing\ both\ A\ and\ B}{Total\ number\ of\ transactions}$$

So, using our example, to calculate the support for the rule {*cheese, eggs*} ⇒ {*bread*} both {*cheese*, *eggs*} and {*bread*{*cheese*, *eggs*} ⇒ {*bread*} ¼ is ¼ or 25%.

This tells us that the rule applies to 25% of our transactions.

The **confidence** of an association rule is the proportion of transactions that contain both *A* and *B*, out of those transactions that we know contain *A*. This is the probability of having *B* in a transaction if we already know we have *A*:

$$confidence(A \Rightarrow B) = \frac{Number\ of\ transactions\ containing\ both\ A\ and\ B}{Number\ of\ transactions\ containing\ A}$$

Using our example again, there are two transactions containing {*cheese*, *eggs*} (transaction numbers 3 and 4) and only one of these also contains {*bread*}, so the confidence for the rule so the confidence for the rule {*cheese, eggs*} ⇒ {*bread*} is ½ 50%.

This tells us that if a transaction has cheese and eggs, there is a 50% probability that it also has bread.

We can see that a rule $A \Rightarrow B$ is more interesting to explore and act upon if it has high support (because there is a high probability that transactions have both A and B) or high confidence (because we know that if a transaction has A, there is a high probability that it also has B), or preferably both. For this reason, when mining for association rules, we often set minimum support and a minimum confidence value so not to spend unnecessary time exploring the association of itemset which does not occur regularly.

The **lift** is a useful way to quantify the strength of the association rule. It compares the probability of obtaining the consequent using the rule (given by the confidence) to the probability of obtaining the consequent in the dataset in general. The formula shows this:

$$lift(A \Rightarrow B) = \frac{confidence\ (A \Rightarrow B)}{percentage\ of\ transactions\ containing\ B}$$

In our example, we already saw that the confidence {*cheese, eggs*} ⇒ {*bread*} is 50%. Only one of the four transactions contains {*bread*}, so the percentage of transactions for {*bread*} is 25%. This gives us a lift for the rule {*cheese, eggs*} ⇒ {*bread*}, which is $\frac{50\%}{25\%}$ = 2.0.

As you can see, we normally write the lift as a number, not a percentage – unlike the support and the confidence, the lift is not a probability but a ratio comparing two probabilities. It can be greater than 1 (as in this case).

The lift of 2.0 tells us that customers who buy {*cheese*, *eggs*} are 2.0 times as likely to buy bread than customers in general. The supermarket may recommend bread with cheese and eggs in their shopping basket to customers when they find this out.

If the lift of a rule $A \Rightarrow B$ is close to 1, this tells us there isn't much difference in the probability of getting consequent *B*, whether or not we have the antecedent *A*. Such rules are not interesting for data mining purposes. We are more interested in rules where the lift is not close to 1, because this

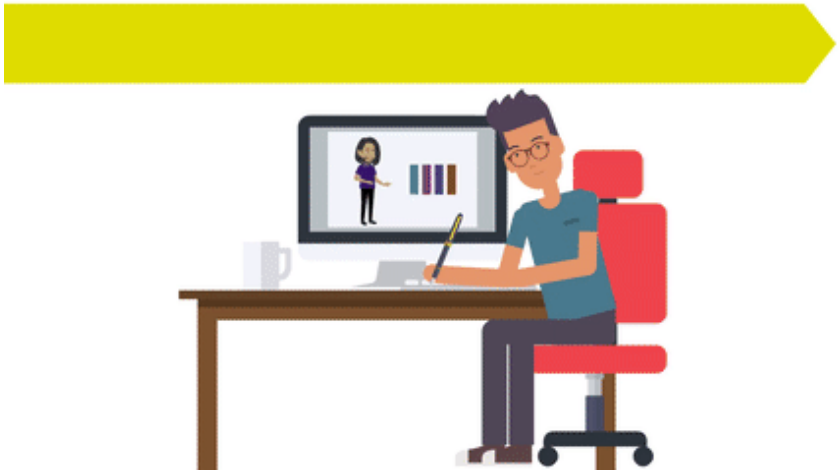
tells us that the fact that a transaction contains *A* appears to influence whether it contains *B* (and the further away the lift is from 1, the greater this influence).

So, to summarise our use of the metrics, we are most interested in rules that have *high support*, *high confidence* and a *lift that is not close to 1*. When data mining for associations, we usually set a minimum level of support and confidence so as not to waste resources over associations that are not going to be interesting. We then calculate the support, confidence and lift of the association rules we have found, so that we can evaluate how interesting or useful the association is.

Continue

Topic 1 - Activity 1

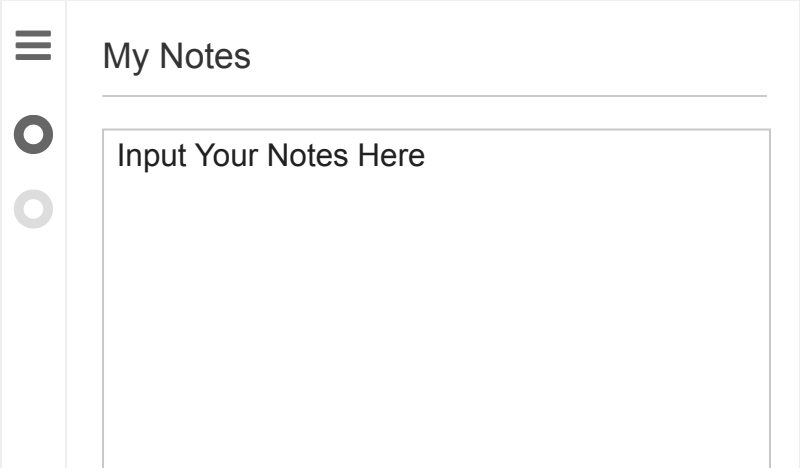
Make a Note



Please make a note listing the key terms defined in this topic in bold and write down their definitions. Write down the formula for the support, confidence, and lift.

You can use the below Notepad to help you with this activity, but feel free to use whichever note-taking tool you prefer.

If you decide to use the below Notepad, please remember anything you type in the box below will not be kept when you leave the lesson, so you will need to download your notes for revision and discussion purposes.



Continue

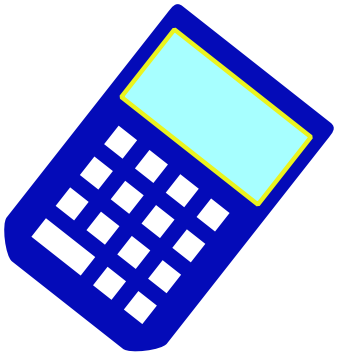
Topic 1 - Activity 2

Calculate the support, confidence and lift for the association rule using the example transactions given in the topic.

$$\{bread\} \Rightarrow \{cheese, eggs\}$$

Also, calculate the support, confidence and lift of the association rule for any two itemsets of your choice – for example, to investigate whether customers who buy milk also buy eggs.

Put your results in the **discussion forum** and **comment** on your findings. Read the results of other people in your class and **comment** on these.



Continue to Topic 2

Topic 2: The apriori algorithm



Image from Articulate Rise 360 Content Library.

In this topic, we look at the apriori algorithm, an efficient algorithm for mining association rules in a large dataset.

Consider, once again, our transactions from the previous topic:

Transaction 1: eggs, milk

Transaction 2: eggs, flour, milk

Transaction 3: bread, cheese, eggs, flour

Transaction 4: cheese, eggs, flour, milk

For the five items sold, there are potentially $2^5 = 32$ possible itemsets (of different sizes). Fortunately, we do not need to check every possible pairing of itemsets for possible associations. For example, we wouldn't {*milk, eggs*} ⇒ {*bread*} because it is not useful to ask whether people who buy "milk and eggs" also buy "milk"! Still, for five items, there are dozens of possible combinations, and it would be too laborious to go through all of them to see if they tell us something interesting about customer habits.

Imagine this was a regular supermarket or an online store with thousands of products. The number of possible combinations is too large to check each one for possible associations.

Note that most combinations of itemsets will have little *support*. There will be relatively few, if any, transactions with these itemsets. One reason for this is that there are certain items that customers tend to buy relatively less often. This means there will be little support for these items alone. If we combine them with other items into an itemset, there will likely be even less support for these.

This is an important property of itemsets that is used in the apriori algorithm, first proposed by Agrawal and Srikant (1994), to make association rule mining more efficient. This is called the "apriori property" and can be stated as follows:

If an itemset Z is not frequent (meaning it has low support), then any other itemsets that contain the items in Z are also not frequent.

The apriori algorithm uses this property to significantly reduce the number of itemsets we need to consider, making the association mining more efficient.

This is done by deciding the minimum level of support required. We then define itemsets that reach this minimum as “frequent itemsets”, and any itemset that does not have the minimum level of support required is “not frequent”. Instead of checking every possible itemset, we will prune the search space by checking only frequent itemsets (because we know in advance that any rules involving the itemsets that are not frequent will not meet the minimum support we are looking for).

We start by finding F_1 (all the *frequent* 1-itemsets, i.e., itemsets with one item). We do this by generating C_1 (all *candidate* 1-itemsets), and then removing any that are not frequent (their support is less than the minimum required). We can then generate C_2(the candidate 2-itemsets, i.e., itemsets with two items) using only items that were in F_1. (This is because we know from the apriori property that if we include any non-frequent items, we will go below the required level of support.) We then remove all the 2-itemsets that do not have enough support from C_2 F_2 (all the frequent 2-itemsets). We carry on generating the next size of C_3, C_4 (etc.) and, each time, we prune out the ones that are below the minimum support level to get the frequent itemsetsF_3, F_4 (, etc) until it is no longer possible to generate more candidates. Let’s use our example transactions to illustrate the algorithm. For this example, we will look for rules with a minimum support of 75%. As there are four transactions in our dataset, we would expect itemsets to occur in at least three transactions to meet this threshold.

First, we generate C_1 (all the candidate 1-itemsets). This consists of all the items in the transactions, so:

$$C_1 = \{bread\}, \{cheese\}, \{eggs\}, \{flour\}, \{milk\}$$

Of these candidates, there is only one transaction with {*bread*} and two transactions with {*cheese*}, so these do not have 75% support. We can take them out, and our frequent 1-itemsets are:

$$F_1 = \{eggs\}, \{flour\}, \{milk\}$$

We now generate C_2 using the items in F_1:

$$C_2 = \{eggs, flour\}, \{eggs, milk\}, \{flour, milk\}$$

The sets {*eggs*, *flour*} and {*eggs*, *milk*} occur in three transactions. So, they meet our minimum support of 75%. But {*flour*, *milk*} only occurs in two transactions, so we can discard it as not frequent. Thus, we get our set of frequent 2-itemsets (F_2):

$$F_2 = \{eggs, flour\}, \{eggs, milk\}$$

Now we generate C_3 from the items in F_2 and we get:

$$C_3 = \{eggs, flour, milk\}$$

The itemset {*eggs*, *flour*, *milk*} contains the subset {*flour*, *milk*} which we discarded before as it only occurs twice; so, using the a priori principle, we can also discard {*eggs*, *flour*, *milk*}.

So, there are no frequent 3-itemsets. As F_3 is empty, we cannot generate any more candidate itemsets.

Putting it all together, we now know that the only itemsets that meet our minimum support of 75% are those in F_1 and F_2 which are:

{*eggs*}, {*flour*}, {*milk*}, {*eggs*, *flour*}, {*eggs*, *milk*}

By combining subsets of these frequent items, we get only four rules that have the minimum support required:

{*milk*} ⇒ {*eggs*}

Note that we have not included rules such as because we already know that the combination {*flour*, *milk*} does not have enough support.

So now, instead of trying all possible combinations of itemsets, we only need to try these four rules. We know that every other possible rule will not have enough support.

We can now calculate the support, confidence and lift for each of these and use the results to understand customer habits.

Continue

Topic 2 - Activity 1

1. The supermarket in our example has five items. Estimate how many possible rules we would need to check if we checked each possible combination. You could do this by attempting a list of the possible combinations, but there are also efficient ways of calculating the number of valid combinations.
2. What is the list of frequent itemsets we would obtain from our example if we used minimum support of 50% instead of 75%? {Hint, there should be four 1-itemsets, five 2-itemsets and two 3-itemsets.)



CONTINUE

Topic 2 - Activity 1

Reading



Pages 213-245 of Chapter 4: Association Analysis: Basic Concepts and Algorithms from Introduction to Data Mining, Global Edition by Tan *et al*.

Continue to Topic 3

Topic 3: Practical implementation of apriori algorithm using Jupyter Notebook



Image from Articulate Rise 360 Content Library.

Now that we have a good understanding of association rule mining, let's put it into practice and do some association rule mining in Jupyter Notebook using the apriori algorithm.

Several Python libraries can perform this algorithm. We will use efficient-apriori, which is available as a package on PyPI and can be installed using the pip command as it is not installed in Jupyter

Notebook by default. The code for this is shown below:

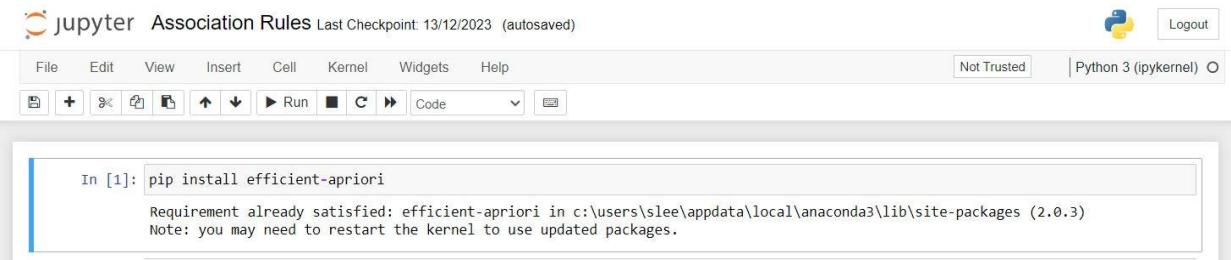


Figure 4.01. Efficient-apriori package installation in Jupyter Notebook by Arden University. (Accessible code).

Once installed, the apriori function can run the algorithm if provided with a list of transactions. The transactions should be presented as a list of tuples or lists, so some pre-processing may be needed to convert the representation of items to this.

When we call the function, we set the minimum support level using the *min-support* parameter.

When it has completed the search, the function returns two values:

- A dictionary containing all the frequent itemsets found for each size (i.e., all the 1-itemsets, 2-itemsets, 3-itemsets, etc.). For each item set, the support count is given.
- A list of rules that were found that meet the minimum support requirement.

Full documentation for this package can be found here: Efficient-Apriori 2.0.3 documentation by Odland.

We will first use the simple example from the previous topic so that you can familiarise yourself with the library, and then we will move on to a large dataset from the real world. For this example, the data set is very small, and we can write it directly into the code as follows:

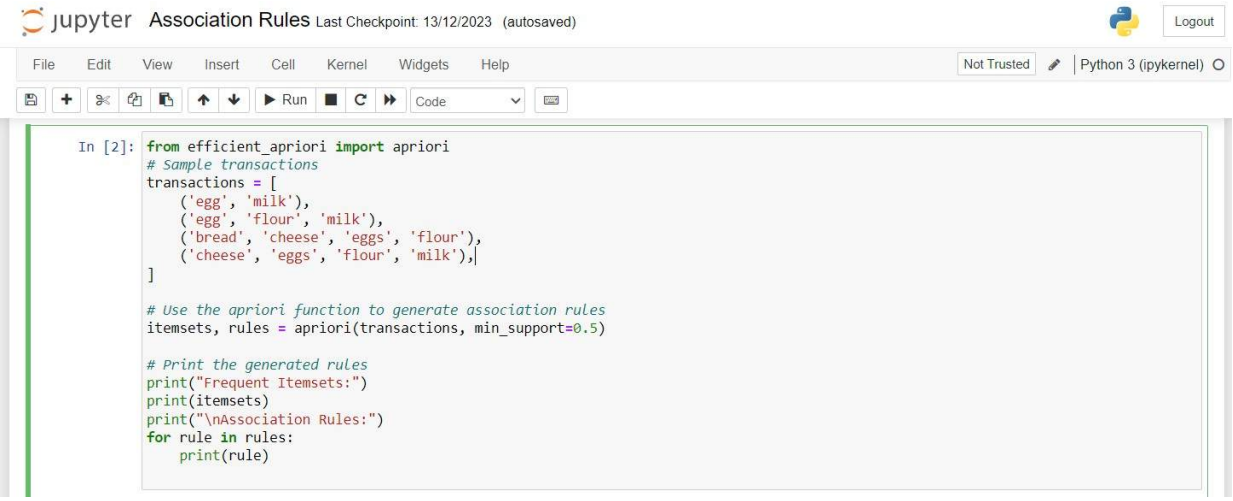


Figure 4.02. Association rule generation in Jupyter Notebook by Arden University. (Accessible code).

The output:

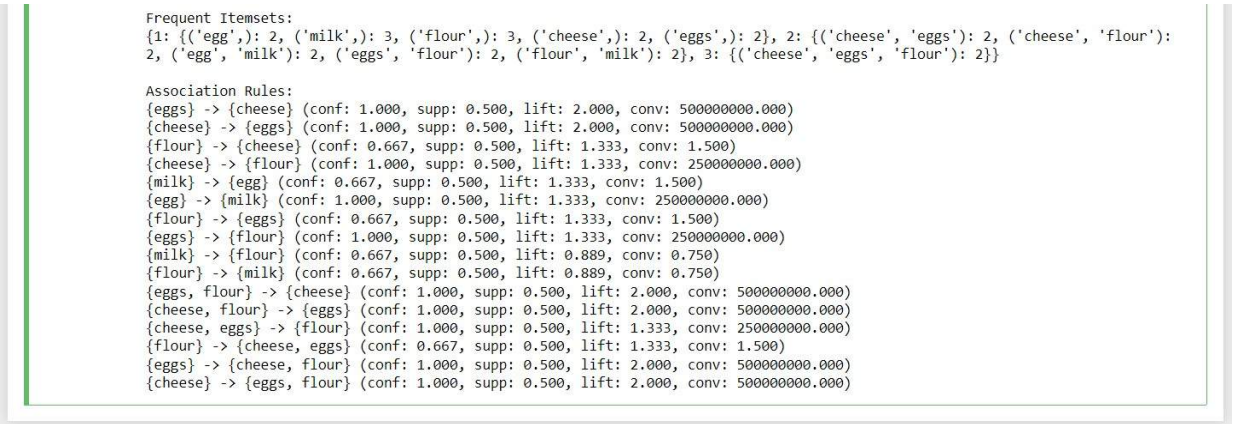


Figure 4.03. Association rules in Jupyter Notebook by Arden University. (Accessible code).

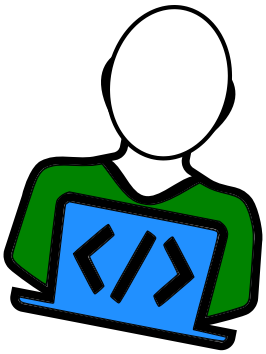
We note that all the rules discovered have a lift above 1.000, telling us that they are not particularly interesting, despite the fact that there is a high level of support and confidence for these rules. This is mainly due to our very small and unrepresentative example dataset. In our dataset, there are only four transactions which all have eggs. Customers who buy cheese, for example, also tend to buy eggs, but this is because everyone buys eggs – customers who buy cheese are not more likely to buy eggs than the customers in general.

Continue

Topic 3 - Activity 1

Implement association rule and explore the following

1. Download “The Bread Basket” dataset into your lesson folder from Bakery Sales Dataset by Kuila.
2. Make sure the efficient_apriori package is installed and imported.
3. Read the bread basket.csv from the Jupyter notebook you created.
4. Implement the association rule model with min_support = 0.01.
5. Print out the rules.



Continue

Topic 3 - Activity 2

Find out what conviction is



Support, confidence, and lift are the main metrics we consider when evaluating association rules. The apriori algorithm also gives us a fourth metric: the **conviction**. Research what the conviction means. Does this change your evaluation of any of the rules you have mined?

Continue

Topic 3 - Activity 3

Reading

Pages 249-274 in Chapter 4: Association Analysis: Basic Concepts and Algorithms from Introduction to Data Mining, Global Edition by Tan et al.



Continue to Lesson summary

Lesson summary

Well done, you have completed Lesson 4.

The lesson started with some theoretical concepts that we need to understand to carry out association rule mining and that will be most useful to organisations. In addition to understanding the purpose of association rule mining and how association rules are written and described, we also looked at the key metrics of association rules: the support, confidence and lift of the association rules. We saw the formulas for calculating these metrics. Remembering these formulas is unnecessary – algorithms will normally calculate the values for us. However, we must understand them to understand the metrics and how to use them.

Here is a summary of these metrics for a general rule $A \Rightarrow B$ and how we use them.

Metric	Description	Use
Support of $A \Rightarrow B$	The proportion of transactions that have both A and B is usually given as a percentage.	If A and B occur together frequently, this indicates that there may be more instances where they occur together in the future, so

Metric	Description	Use
	We also sometimes talk of the *support count*, which is the actual number of transactions with both A and B.	acting on this would be beneficial. For this reason, the higher the support, the more interesting the rule is.
Confidence of $A \Rightarrow B$	The probability that a transaction contains B, if we know that it contains A. Usually given as a percentage.	The higher the confidence, the more confident we can be that if, in future, a transaction contains A it will also contain B. So, the higher the confidence, the more interesting the rule is.
Lift of $A \Rightarrow B$	If the lift is x, this means that transactions that contain A are x times more likely	If the lift is close to 1, there isn't much difference to how likely we will have B or whether we have A. We are more interested in rules

Metric	Description	Use
	to contain *B* than the transactions in general. This is given as a number.	where the lift is far from 1.

We then looked at the apriori algorithm, which is a commonly used algorithm for finding association rules because it makes the search efficient by eliminating itemsets that are not frequent enough to be worth considering. We gained an understanding of why it is efficient using a small example – but, once again, when we are doing association rule mining on a large dataset, we would not need to implement the algorithm from scratch as it is built into libraries that can be used on all platforms such as Jupyter Notebook. This means the analyst can focus on interpreting the results.

This is exactly what we did with a practical example using real data. Like the other lessons in this module, the examples give you a starting point, which you can implement and explore to increase your familiarity with the tools.

Continue to Knowledge check

Knowledge check

Select the correct answers for the following questions.

1 of 2: In a market basket analysis, the rule {*Pizza*} ⇒ {*Beer*} has a lift of 3.0. What does this mean?

- ◯ People who buy pizza are three times more likely to buy beer than the customers who do not buy beer.
- ◯ People who buy pizza are three times more likely to buy beer than customers in general.
- ◯ People who buy beer are three times more likely to buy pizza than customers in general.
- ◯ There are three times more customers who buy beer than pizza.

SUBMIT

2 of 2: Here are the metrics of some association rules. Which of them is the most interesting?

- Support = 0.21, Confidence = 0.73, Lift = 1.8
- Support = 0.21, Confidence = 0.73, Lift = 1.1
- Support = 0.21, Confidence = 0.5, Lift = 1.8
- Support = 0.14, Confidence = 0.73, Lift = 1.8

SUBMIT

Continue to Key reading

Reading List

All reading and resources in this module are provided by the Arden University Library service. The full reading list for this module can be accessed at the link below. More books, journals and databases can be found on the Library Portal. If you need these resources in a different format, our accessibility tools can help.

Full reading list

You can find all the key, essential and wider reading and resources for this module on the dedicated reading list page on the Learning Success Hub.

VIEW READING LIST

References

Cited sources that do not appear in your reading list:

Agrawal, R. K. and Srikant, R., (1994). Fast Algorithms for Mining Association Rules in Large Databases [online]. *In:* Bocca, J. B., Jarke, M. and Zaniolo, C., eds. *20th International Conference on Very Large Data Bases*, Santiago de Chile, Chile, 12-15 September 1994. Burlington, MA: Morgan Kaufmann Publishers.

Cheng, C. K., Paré, D. E., Collimore, L.-M. and Joordens, S., (2011). Assessing the effectiveness of a voluntary online discussion forum on improving students' course performance. *Computers & Education* [online], 56 (1), pp. 253-261.

MacKenzie, I., Meyer, C. and Noble, S., (2013). *How retailers can keep up with consumers | McKinsey* [online]. New York, NY: McKinsey & Company.